Marketing Campaign Response — Summary

Predicting who responds to a campaign, and turning it into a targeting and budget plan.

▶ **Live interactive app:** https://marketing-campaign-response-ajrjqsucchbawv2ykmcklt.streamlit.app

Full Jupyter notebook: view the complete analysis & code

2,237 CUSTOMERS	**14.9%** BASELINE RESPONSE	**XGBoost** BEST MODEL	**0.90** ROC-AUC	**~3×** WARM-LIST LIFT

Bottom line. A retailer's last campaign reached everyone, but only ~15% responded, so most of the spend was wasted. The model scores each customer's likelihood to respond (ROC-AUC ≈ 0.90). Aimed at the right people, the same budget earns far more.

Approach

Cleaned 2,237 customer records (filled 24 missing incomes, removed a few typo rows, tidied messy labels), then built 10 behaviour features: spending, recency, engagement, past-campaign history, income tiers. Trained and compared three models inside leak-free scikit-learn pipelines with 5-fold cross-validation. Because only ~15% respond, models are judged on ranking quality (ROC-AUC) and recall, not plain accuracy, and each is told to weight the rare "yes" cases.

Key insights

- **Past responders are gold:** people who accepted a past campaign respond at **41%** vs **8%** (~3×).
- **Higher income, higher response: 28%** vs **10%**; high income + a past acceptance reaches **47%.**
- **Recent buyers are warm:** responders last bought ~35 days ago vs ~52 for everyone else.
- **Households without kids respond more: 27%** vs **10%** with children.
- **Catalog buyers over-respond** (4.2 vs 2.4 purchases); website *visits* don't predict response at all.

Model performance

Model	Acc	Prec	Rec	F1	AUC
XGBoost ★	**0.89**	**0.63**	**0.61**	**0.62**	**0.90**
Random Forest	0.89	0.72	0.42	0.53	0.89
Logistic Reg.	0.81	0.42	0.73	0.53	0.88

XGBoost wins on ranking (ROC-AUC) and F1. Since a missed responder (~$11) costs more than a wasted contact (~$3), we lean toward recall and set the contact cutoff where expected profit is highest, not at a textbook 0.5.

Business recommendations

- **Contact first:** rank everyone by predicted probability; start with the ~460 prior responders (~3× the base rate, free to find).
- **Budget:** concentrate on the top 2–3 probability deciles; stop funding the unresponsive bottom half.
- **Personalise:** premium offers for high-income responders, recency-triggered sends for warm buyers, sustained catalog investment.
- **Prove it:** track profit per contact against a random-targeting control so the lift is provable in dollars.

What drives a response

The model's top factors, in order: **past campaign acceptances** (by far the strongest), the **recency** of the last purchase, **customer tenure**, then **income and spending** (especially wine, meat, and catalog buying). Being single or highly educated nudges response up; children at home pull it down. In one sentence: the most likely responder is a **recent, comfortable, previously responsive buyer with no kids at home**, and the model puts a number on each trait.

Deployment risks & what's next

Watch for: wasted spend from false positives (cap contact frequency, tune the cutoff), missed sales from false negatives (lean toward recall, keep a small random holdout), and data drift as customer behaviour shifts (monitor the inputs, retrain on a rolling quarterly window). **Next steps:** customer segmentation to pair the right offer with each group, an A/B test against random targeting to prove the lift in dollars, and probability calibration so the scores can be trusted as money.

Expected business impact. The same budget, aimed at customers who respond about three times more often, turns into materially more accepted offers per dollar and far less wasted contact spend. More than a one-off model, it gives the marketing team a repeatable, measurable way to decide where the next dollar earns the most — targeting instead of broadcasting, with the lift provable against a control group.

Built with Python, pandas, scikit-learn, XGBoost, and Streamlit. The full method, code, and charts live in the accompanying Jupyter notebook (linked above).